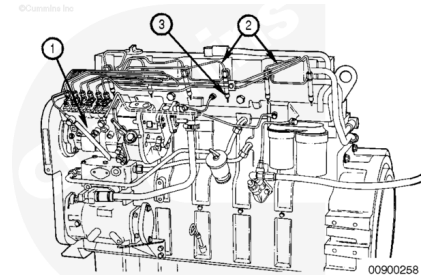


Trainings ▾

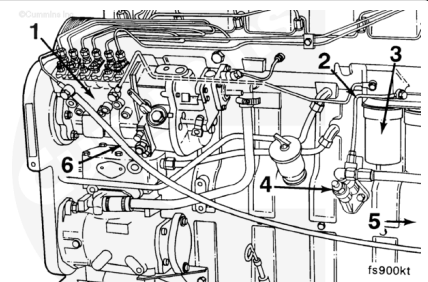
General Information

The function of the fuel system is to inject a metered quantity of clean atomized fuel into the engine cylinders at a precise time near the end of the compression stroke. The components of the fuel system contribute to the delivery of fuel to the cylinders.

1. Fuel injection pump
2. High-pressure fuel lines
3. Injectors.

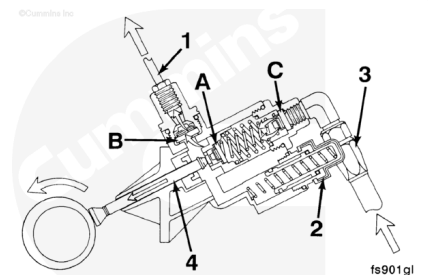


1. Fuel injection pump
2. Fuel supply line
3. Fuel filter
4. Fuel transfer pump
5. Fuel tank (**not** shown)
6. Fuel return line.

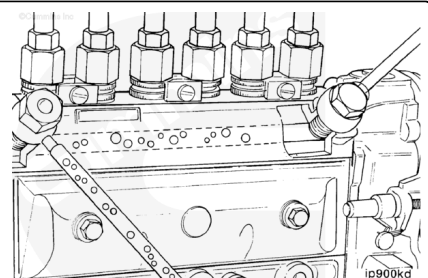


The fuel transfer pump is mechanically driven by a plunger running against a special lobe on the camshaft. The fuel transfer pump contains a pumping piston (A) and check valves (B) (C) to control the flow of fuel and bleed back during engine shutdown.

1. Low-pressure supply line
2. Priming pump
3. Fuel inlet line
4. Plunger.



The pressure relief valve arrangement in the supply side of the fuel circuit creates a self-bleeding system on the A-type fuel injection pump. Air introduced during replacement of any supply-side components will automatically bleed out of the fuel system.



Small amounts of air can be bled from the fuel injection pump by operating the hand primer on the fuel transfer pump or by cranking the engine.

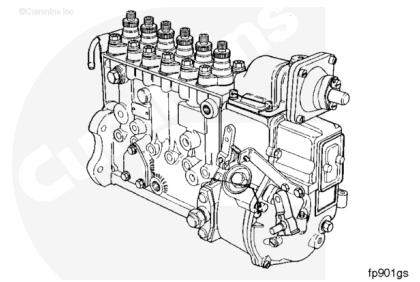
Marine 480C-E engines use a Bosch® P7100 mechanical fuel pump with a Bosch® RE30 governor. A mechanical, positive displacement fuel lift pump provides pressurized fuel to the fuel filter head inlet. A keyswitch-activated electric fuel shutoff valve is mounted on the outlet of the fuel filter head. Filtered and pressurized fuel is then routed back to the fuel pump inlet.

The purpose of the pressure relief system installed on the lift pump is to prevent over-pressurizing of the fuel filter assembly upon shutdown. When the engine is commanded to shut down, the keyswitch circuit is interrupted and causes the fuel shutoff valve to close. As the engine comes to a stop, fuel is still pumped to the fuel filter from the lift pump. It is possible that the pressure can climb high enough to cause the fuel filter seal or body to fail.

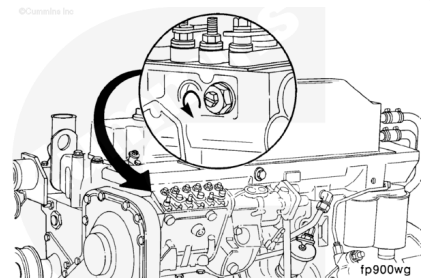
The system uses a relief valve in the banjo connection screw located at the outlet of the lift pump. A hose is connected to the banjo to direct the fuel back to the lift pump inlet. The relief valve is a ball and spring type and is set to open at 690 kPa [100 psi] +/- 70 kPa [10 psi] and is **not** adjustable. The relief valve should be open **only** after the engine has been commanded to stop.

If the relief valve becomes stuck or does **not** seat completely, fuel will continuously be bypassed back to the lift pump inlet. This causes low lift pump output pressure and leads to poor performance and hard starting.

Following the appropriate Troubleshooting Symptom charts in the Troubleshooting and Repair Manual will identify low lift pump pressure associated with a failed relief valve.

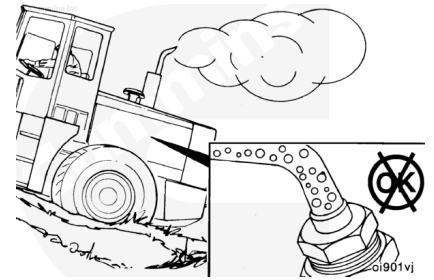


Note : MW-type fuel injection pumps without the fuel drain arrangement on the engine side will require additional venting prior to initial start-up, fuel injection pump replacement, or if the engine has been allowed to run out of fuel. Refer to Procedure 006-003 (/qs3/pubsys2/xml/en/procedures/41/41-006-003.html) for more information.

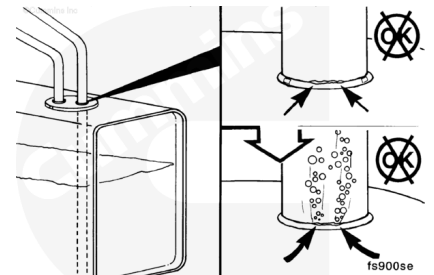


Air from uncorrected leaks in the supply circuit will make the engine:

- Difficult to start
- Run rough
- Misfire
- Produce low power
- Emit excessive smoke
- Produce fuel knock.



A source, often overlooked for air to enter the fuel system, is between the inlet of the prefilter and the suction tube in the tank. Fuel tanks that have the outlet fitting at the top will have a suction tube that extends to the bottom of the tank. Cracks or pin holes in the weld that joins the tube to the fitting can allow air to enter the fuel system.



Since the fuel transfer pump provides positive pressure through the fuel filter and supply line to the fuel injection pump, loose connections or defective seals will show as a fuel leak.

